Investigating Reflection

Exam code: 0625 & 0972

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Investigating reflection

Extended tier only

Aims of the experiment

- To investigate reflection by a Plane mirror

Variables

- **Independent variable** = angle of incidence, *i*
- **Dependent variable** = angle of reflection, *r*
- Control variables:
  - Distance of ray box from mirror
  - Width of the light beam
  - Same frequency / wavelength of the light

Equipment list

Equipment	Purpose
Ray Box	To provide a narrow beam of light that can be easily reflected
Protractor	To measure the angles of incidence and reflection
Sheet of Paper	To mark the lines indicating the incident and reflected rays
Pencil	To draw the incident and reflected ray lines onto the paper
Ruler	To draw the incident and reflected ray lines onto the paper
Plane mirror	To reflect the light beam

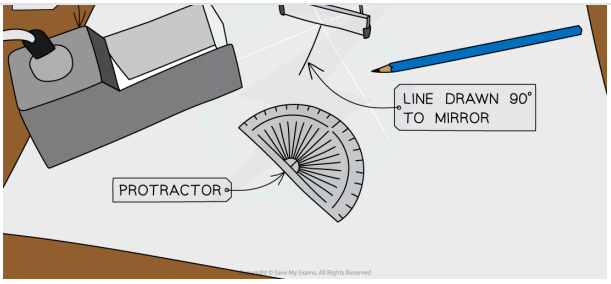
Method

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Apparatus to investigate reflection

1. Set up the apparatus as shown in the **diagram**
2. In the middle of the paper use a ruler to mark a straight line about **10 cm** long
3. Use a protractor to draw a **90° line** that bisects (cuts in half) the 10 cm line
4. Place the mirror on the **first line** as shown in the diagram above
5. Switch on the **ray box** and aim a beam of light at the point where the two **drawn lines cross** at an angle
6. Use the pencil to mark **two positions** of the light beam:
   - A point just after leaving the ray box
   - The point on the reflected beam about 10 cm away from the mirror
7. Remove the ray box and mirror
8. **Use a ruler t**o join the two marked positions to the point where the originally drawn lines crossed
9. Use the protractor to measure the two angles from the 90° line. The angle for the ray towards the mirror is the angle of incidence, and the other is the angle of reflection
10. **Repeat** the experiment three times with the beam of light aimed at different angles

Results

Example results table

Angle of incidence, *i* (°)	**Angle of reflection, *r* (°)**
10	
30	
45	
80	

Analysis of results

- The law of reflection states:

$$i = r$$

- Where:
  - i = angle of incidence in degrees (°)
  - r = angle of reflection in degrees (°)

Evaluating the experiment

Systematic errors

- An error could occur if the 90° lines are drawn incorrectly
  - Use a set square to draw perpendicular lines
- If the mirror is distorted, this could affect the reflection angle, so make sure there are little to no blemishes on it

Random errors

- The points for the incoming and reflected beam may be inaccurately marked
  - Use a sharpened pencil and mark in the middle of the beam
- The protractor resolution may make it difficult to read the angles accurately
  - Use a protractor with a higher resolution

Safety considerations

- The ray box light could cause **burns** if touched
  - Run burns under cold running water for at least five minutes
- Looking directly into the light may **damage the eyes**
  - Avoid looking directly at the light
  - Stand behind the ray box during the experiment
- Keep all **liquids** away from the electrical equipment and paper
- Take care using the **mirror**
  - Damages on the mirror can affect the outcome of the reflection experiment

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